

Health vs. Price

Testing whether eating healthy really costs more?
using openfoodfacts U.S. grocery data

BA780 Team 2

Aashna Soni
Alex Chen
Dhruv Thakkar
Gauri Nandihalli
Peter Chang

Does eating healthy really cost more?

The Problem

Can shoppers eat healthier without paying more, and where does their grocery money actually go?

Why it Matters

Budget shoppers are told healthy food is expensive, so many stop trying. If that belief is wrong, they can eat better and save.

Our Approach

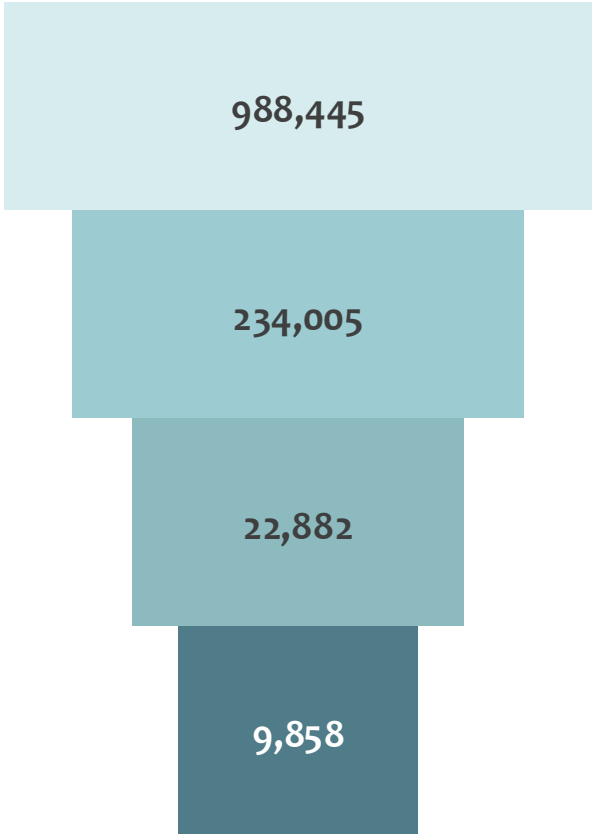
Match products to shelf prices by barcode, compare price per 100 g within same aisle, test 8 hypotheses to find system to suggest healthier swaps.

About the Open Food Facts dataset

- A free and crowd-sourced database of packaged foods like a Wikipedia for food labels.
- Volunteers scan barcodes and photograph labels.
- Each product carries its nutrition per 100 g plus three scores:
  - Nutri-Score (health)
  - NOVA (processing)
  - Green-Score (environment)

Its sister project, Open Prices, collects photographed shelf prices.

How 988,445 products narrowed to the ones we could price and analyze



All U.S. products
From Open Food Facts after cleaning

Products with Categories
After joining with Food Groups data to get categories

Products with a Shelf Price
After joining with Prices

And a known pack size
Products have pack sizes to compute price per 100g for comparisons

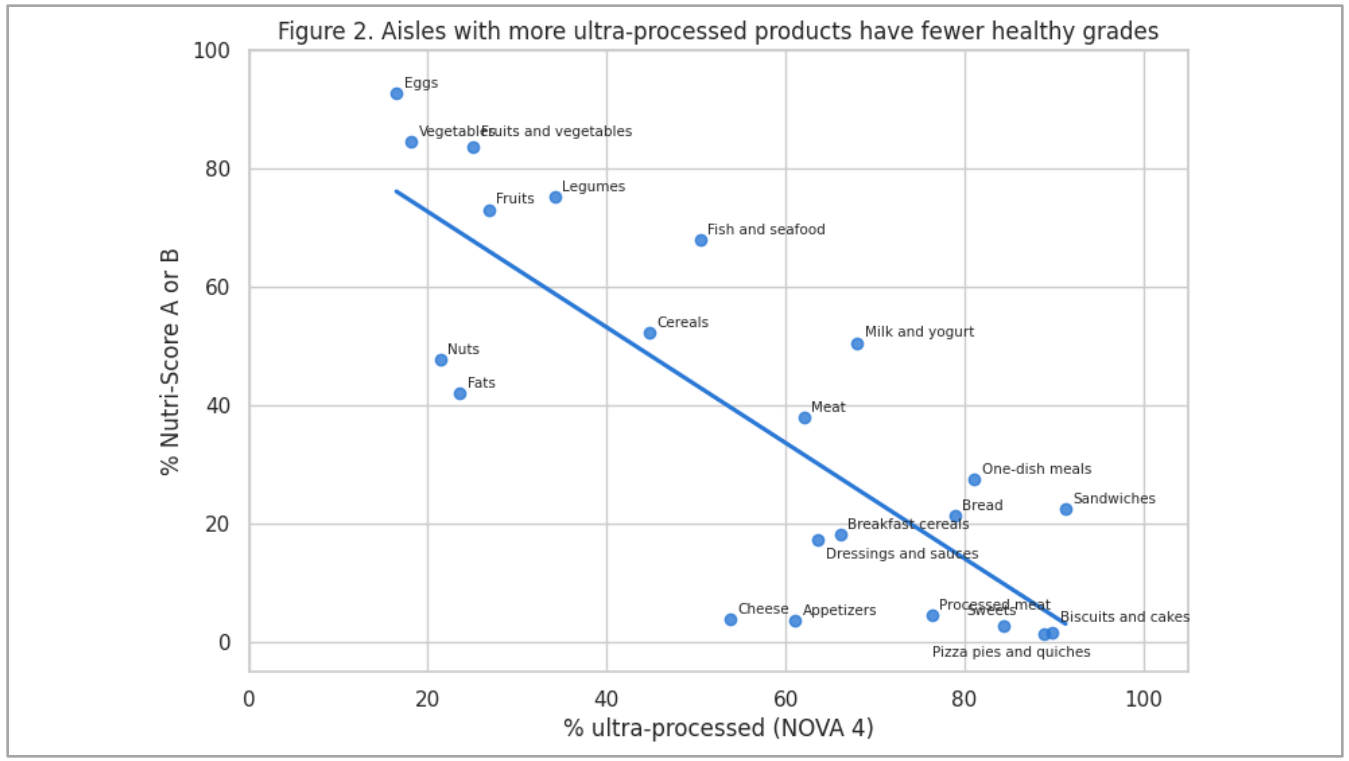
Data coverage for key metrics

What it tells us	Open Food Facts column	Share
Nutrition per 100 g	sugars_100g, fat_100g …	68-85%
Health grade (Nutri-Score)	nutriscore_grade	39%
Processing (NOVA)	nova_group	40%
Aisle (food group)	food_groups_tags	29%
Pack size	size_g_ml	18%

What does it mean to shop healthy?

More ultra-processing goes with fewer healthy products

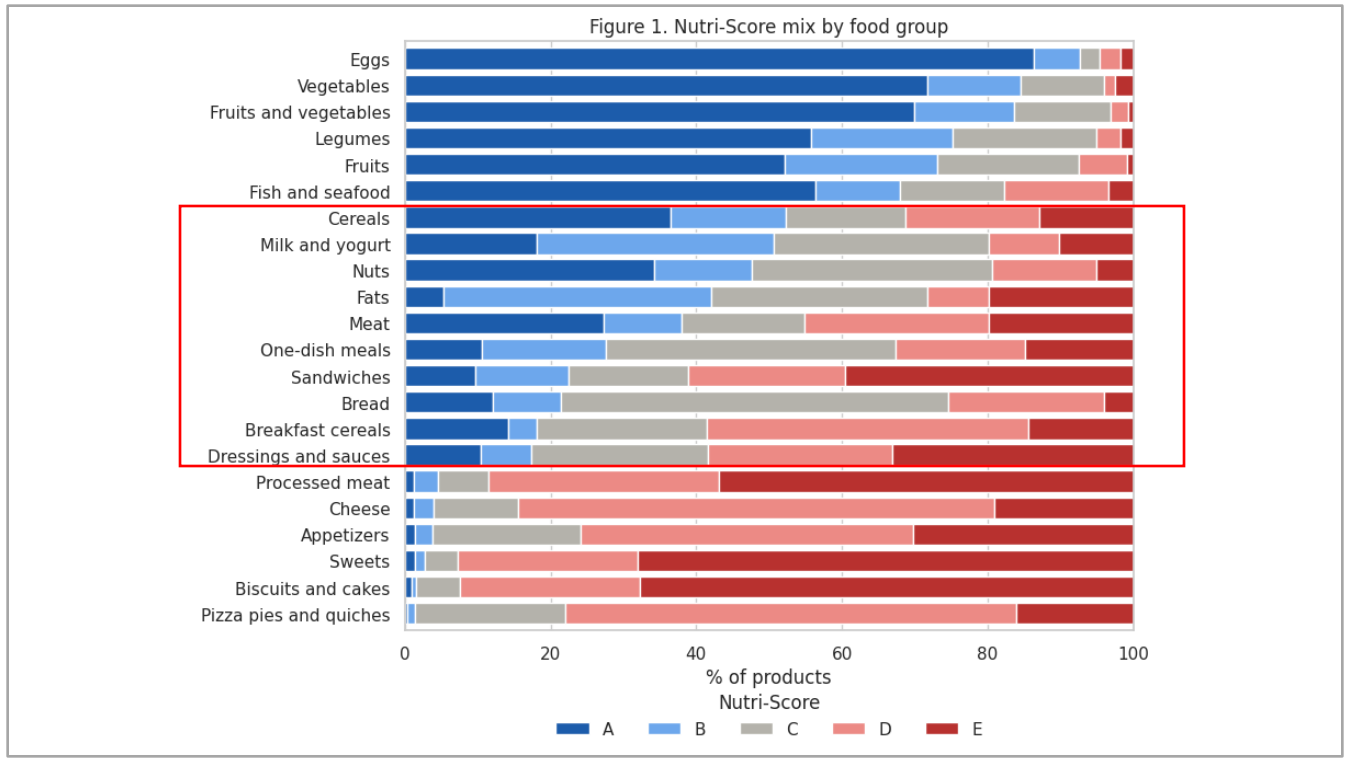
The more ultra-processed an aisle, the fewer healthy products it has



- Strong negative correlation: Spearman ρ = −0.77
- Food groups with more NOVA 4 products generally have fewer Nutri-Score A/B products.
- Exceptions show that processing and nutrition are not the same measure.
- Takeaway: Check both NOVA and Nutri-Score.

For most aisles there's a mix of nutritional value in products and making healthier choices becomes even more necessary

Nutri-Score mix inside each food group

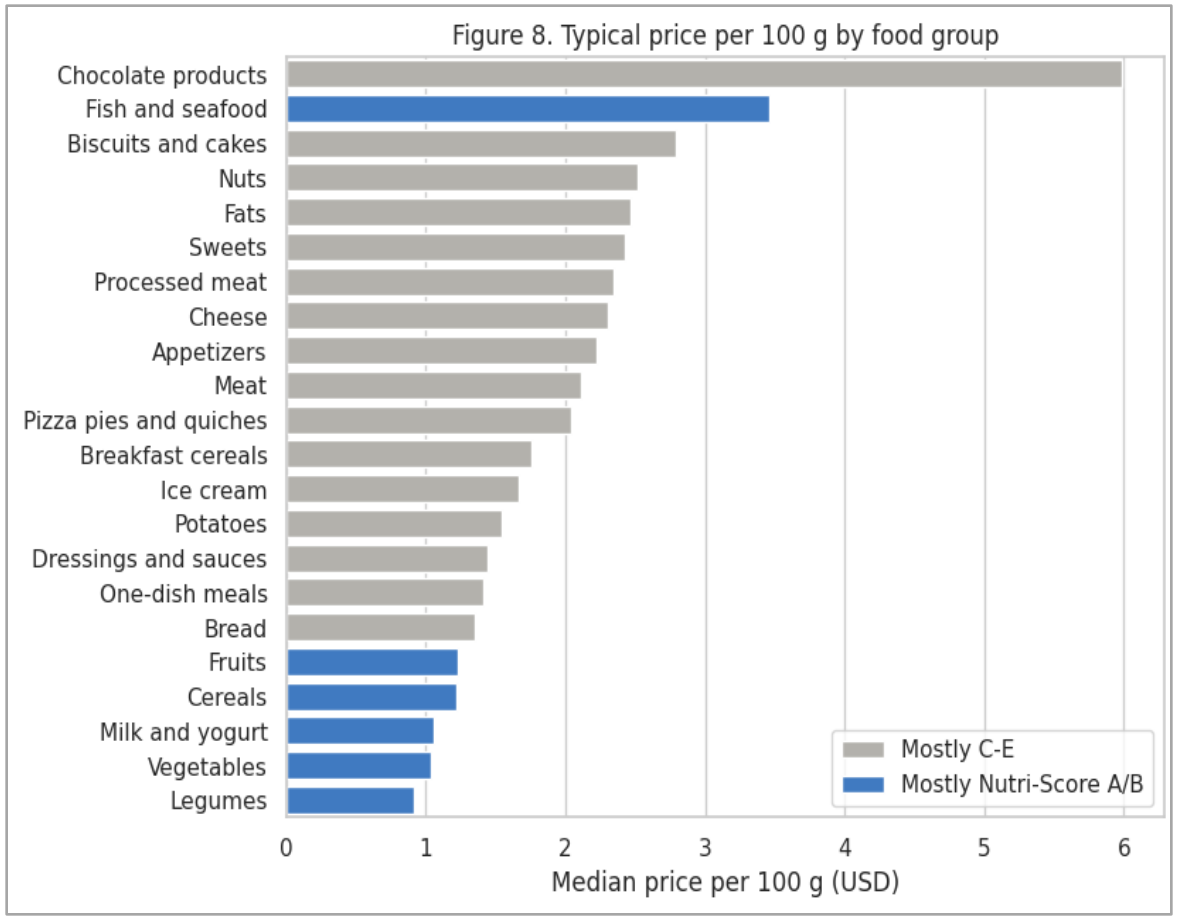


- 93% of eggs and 85% of vegetables score A/B.
- Only 2% of biscuits and 3% of sweets score A/B.
- Cereals, meat and ready meals have a mix of nutritional grades.
- Takeaway: Compare nutrition labels within each food group.

Do customers need to pay a price premium to eat healthier?

Across the store, the healthiest aisles are the cheapest

Are the healthier aisles the more expensive ones?
(median price per 100 g by food group)



≈$1 (healthy)

per 100 g

≈$2-6 (processed)

per 100 g

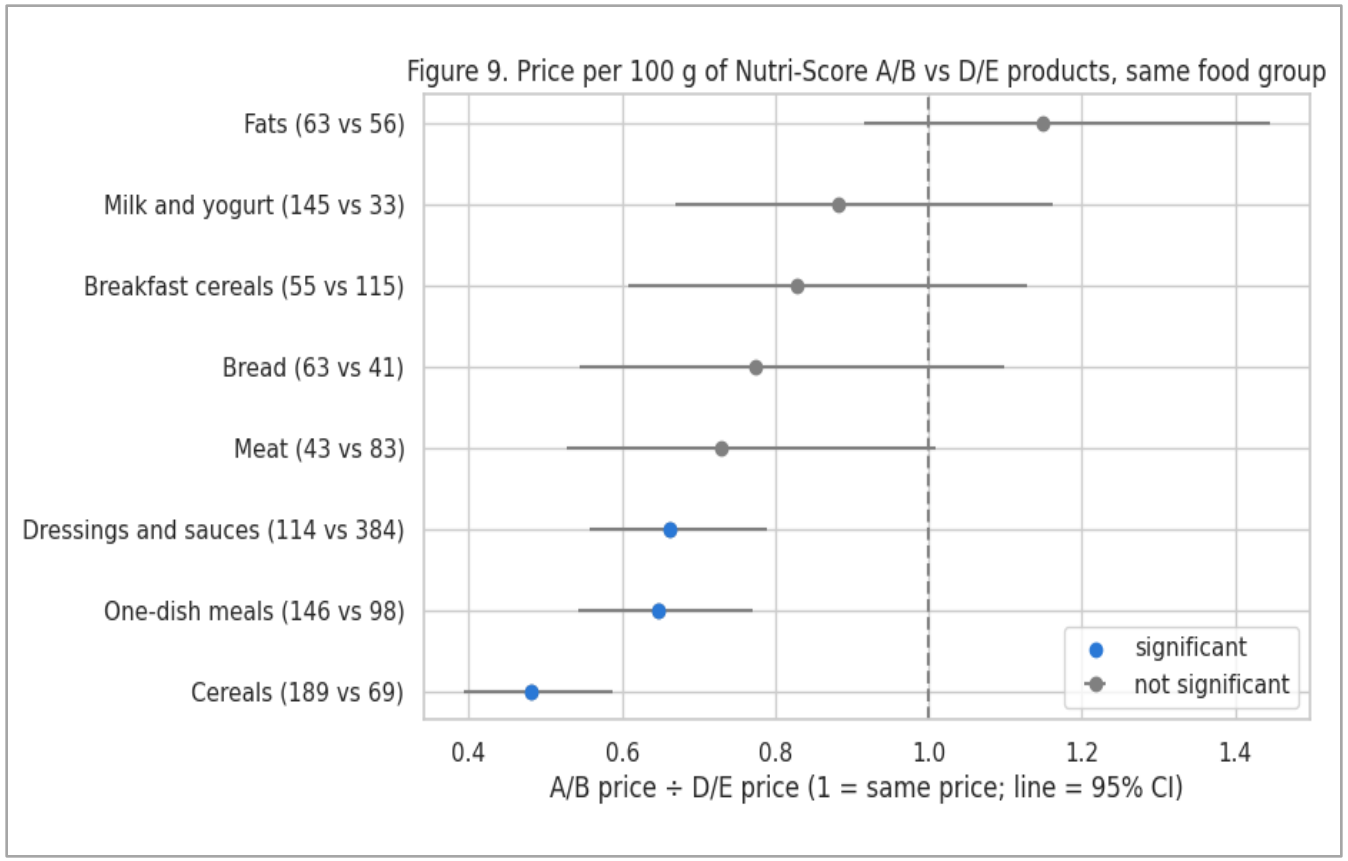
Key takeaways

Healthy aisles are cheapest

"Healthy is expensive" is a pricing illusion:
It shows up per package, not per 100 g.

Within an aisle, healthier products cost less per 100 g

Inside a food group, do Nutri-Score A/B products cost more per 100 g than D/E products?



17% Cheaper

A/B vs D/E price per 100 g, same aisle (95% CI 0.79–0.88)

Key takeaways

Healthy aisles are cheapest

No aisle charges a significant health premium:
Swapping to healthier grade saves money

Promotions: On sale doesn't mean a better deal

Do promotions reward healthy or unhealthy choices?



Same 24%

average discount at every processing level

Key takeaways

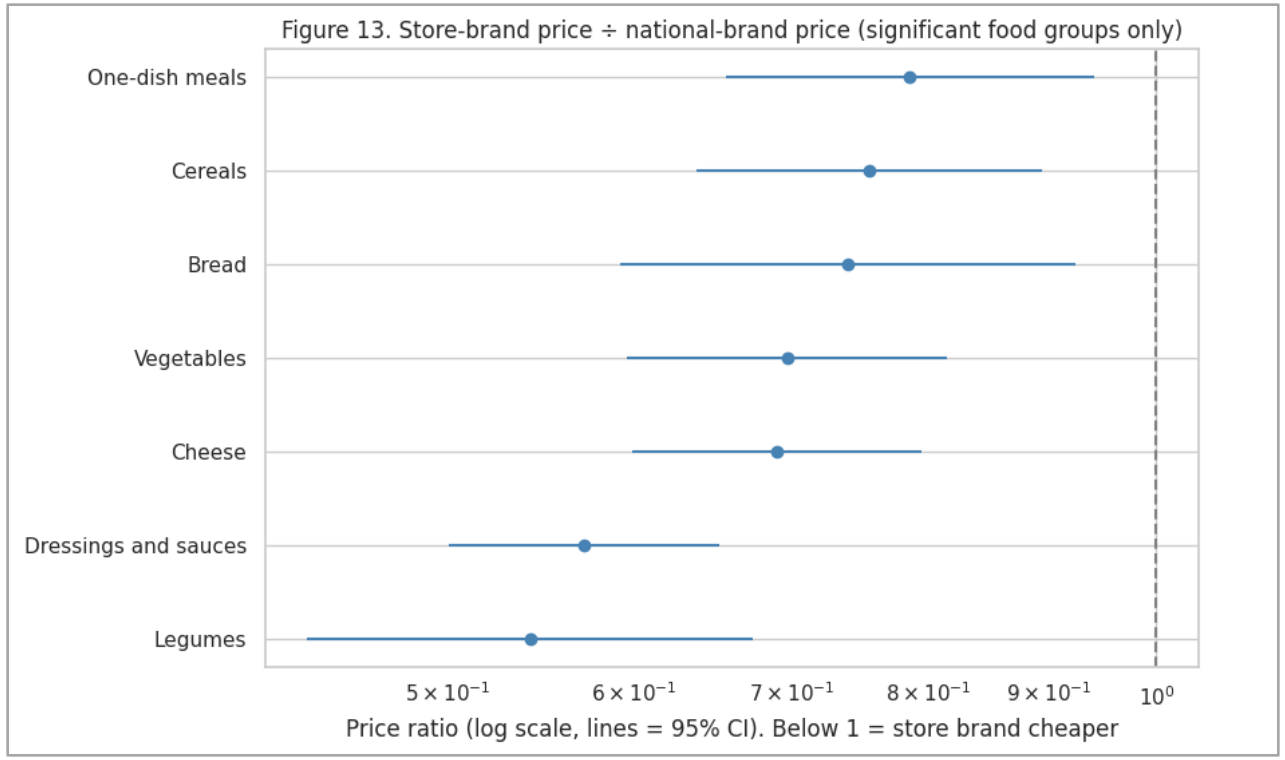
Snacks get more sale tags (+2.4 pp), not bigger discounts (24%)

Even at 24% off,
Snacks cost more than Healthy Options
$1.75 vs $1

Store Level Analysis

Store brands are the biggest saving we found

Store brands vs national brands in the same food group

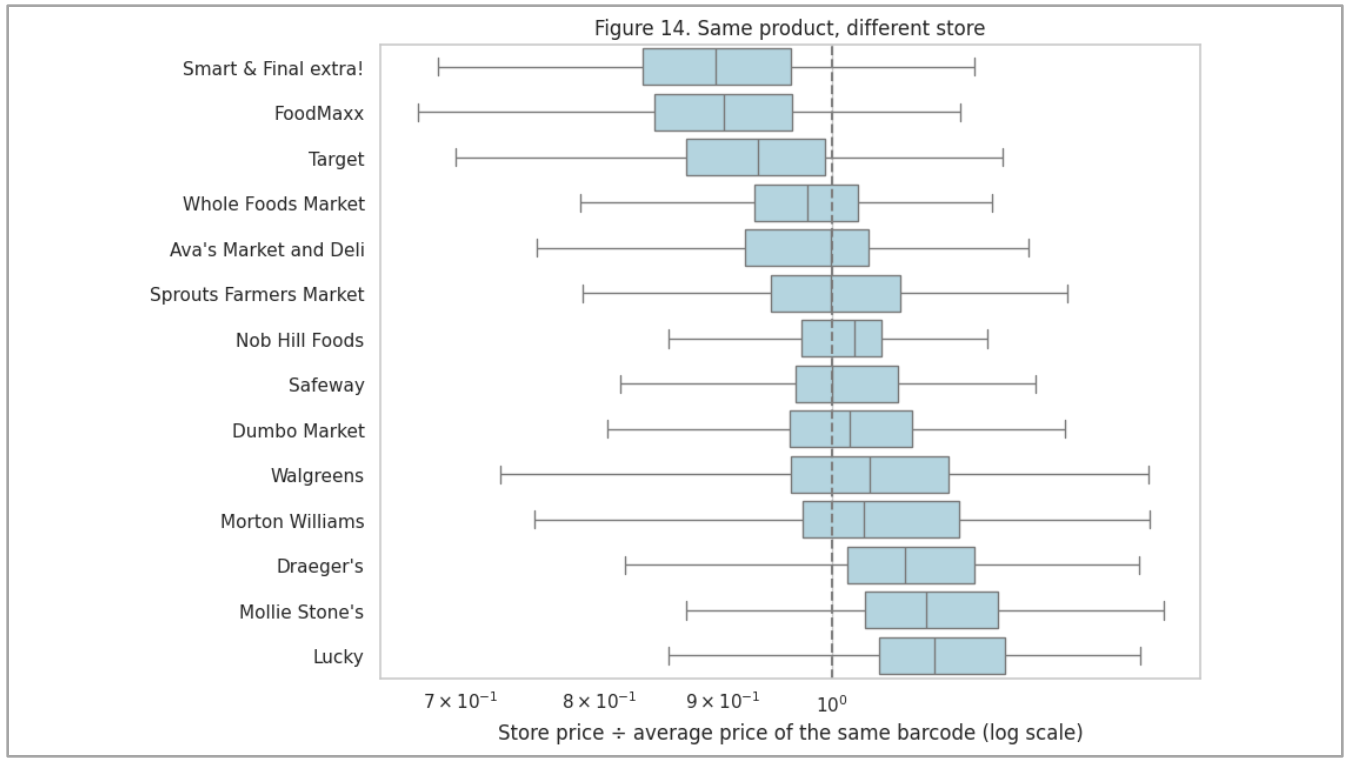


Key takeaways

- Store brands 0.58× cheaper than national brands per 100 g in the same aisle.
- Store brands also have 11 points more nutri-score products and 23 points less ultra-processed.
- Part of the gap is bigger packs and plainer products, which is good news for shoppers rather than a catch. Trying the store brand first is the single easiest rule to follow.

The same product can cost 14% more at a different store

The same product (same barcode) at different stores



Key takeaway

- priciest vs cheapest store for the median product (4,323 products at 2+ stores)
- The same barcode varies 14% between the cheapest and priciest store for the median product 29% or more for a quarter of products.
- Smart & Final and FoodMaxx are about 11% below average, while Whole Foods is slightly below average for identical items: its expensive reputation comes from what it stocks, not what it charges. Where you do the big weekly shop matters as much as what you buy.

The smart-shopper playbook: How you compare matters more than how much you pay

The aisle is the baseline, the real choice happens inside of it.

Nutrition varies within the aisle.

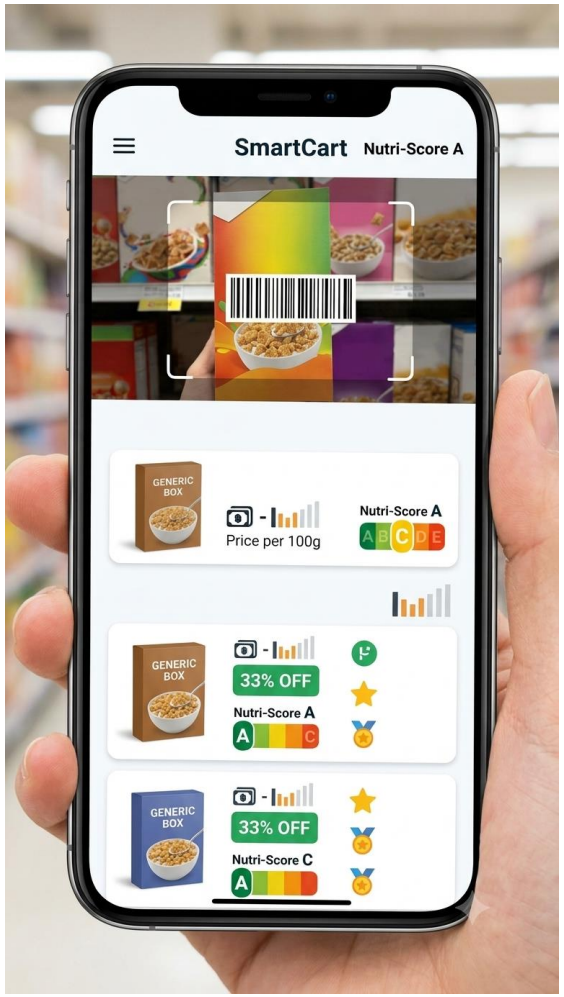
Healthier food doesn't necessarily mean higher prices.

The healthier options in the same aisle tend to cost less.

Save by changing where and what brand you buy.

Stores and brands decide how much you pay.

SmartCart: a gamified swap app built on Open Prices



What the shopper gets

- Scan and price check vs the aisle median
- Up to 3 healthier, cheaper swaps
- Store tip for the same barcode
- Points, streaks and badges
- Points for photographing price tags, which feed Open Prices

Rule based swap based on Open Food Facts data



Scanned | Nutri-Score C

Cinnamon Toast Crunch

$2.05 per 100g
1.2 ×the aisle median



Swap | Nutri-Score A

Post Shredded Wheat

33% cheaper per 100 g

Available data largely limits the analysis

Local, concentrated price data

85% of price observations are from California and 93% from one contributor, so results describe these stores, not U.S. prices.

Small priced sample

5,752 foods have both a price per 100 g and a food group, and package size is known for only 43% of priced products.

Scores are not missing at random

popular, well-documented products are more likely to have a Nutri-Score, Green-Score, and price.

Correlations does not equal to causations

we show which products cost less, not why. Brand, packaging, and store strategy all play a part.

Thank you!

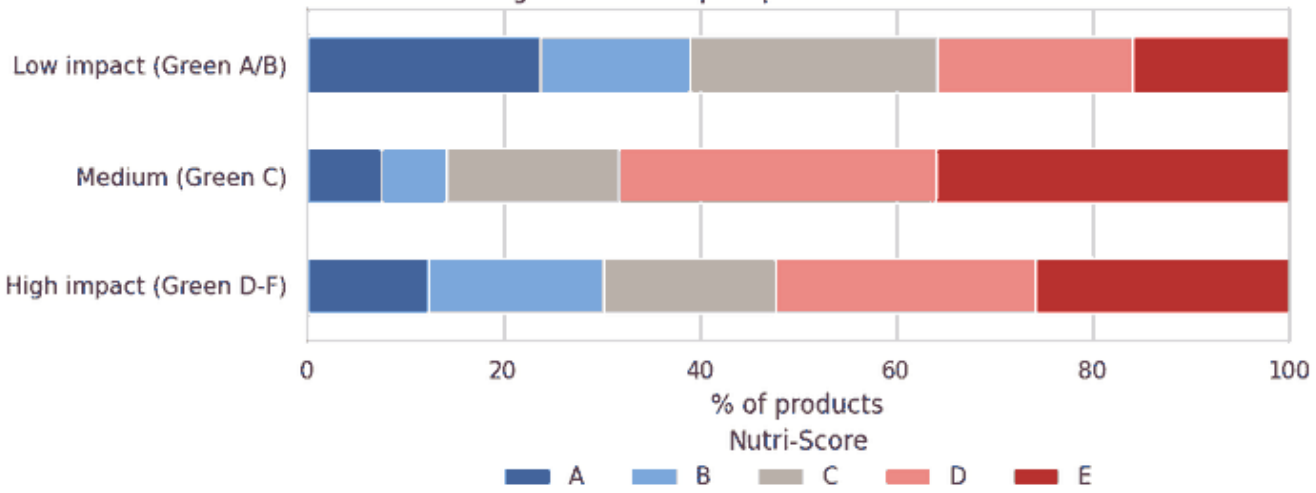
Appendix

Supporting charts from the notebook for Q&A

Low-impact products are only slightly healthier overall

Low-impact products are only slightly healthier across the whole store

Figure 3. Low-impact products are a bit healthier



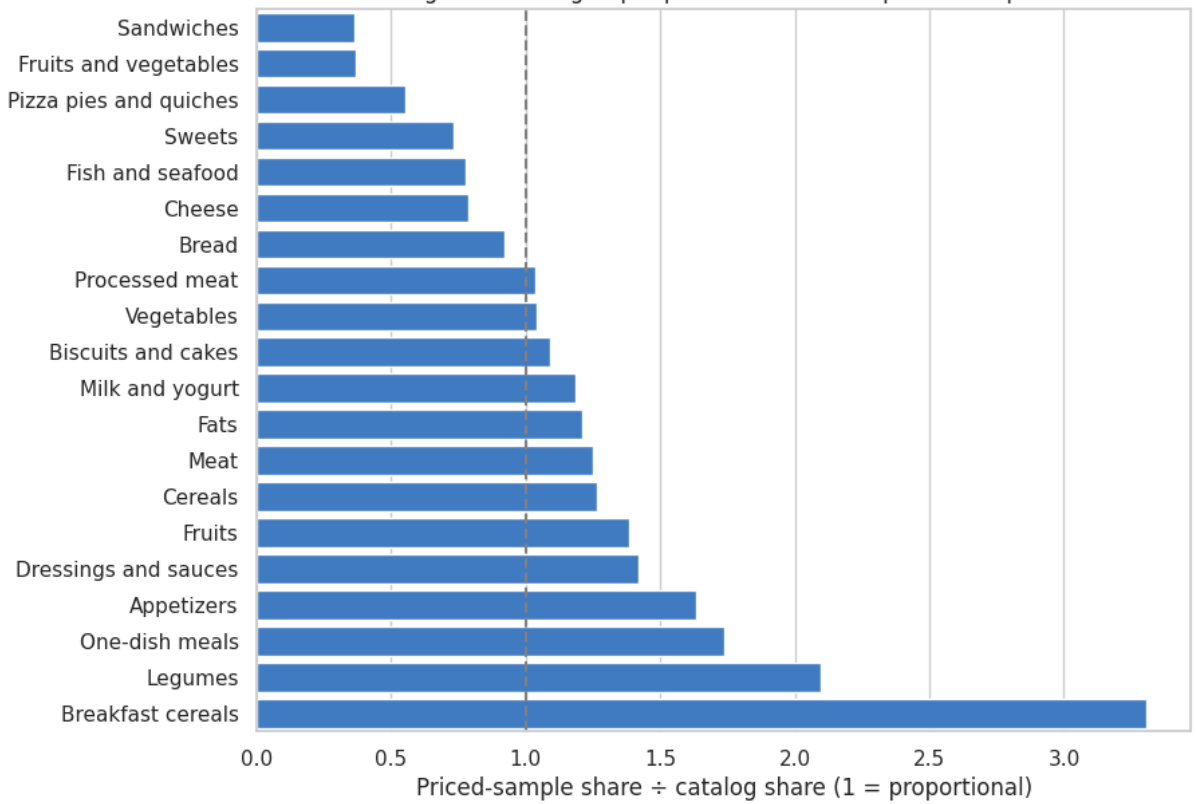
Key takeaway

Low-impact products are only slightly healthier (39% score A/B vs 30% for high-impact), and medium-impact products are the least healthy (14%). Therefore, Green-Score alone is a weak signal of health, so we should compare products within the same food group next to explore more meanigful links.

The priced sample mostly mirrors the full catalog

Is each food group over- or under-represented in the priced sample?

Figure 7. Food group representation in the priced sample

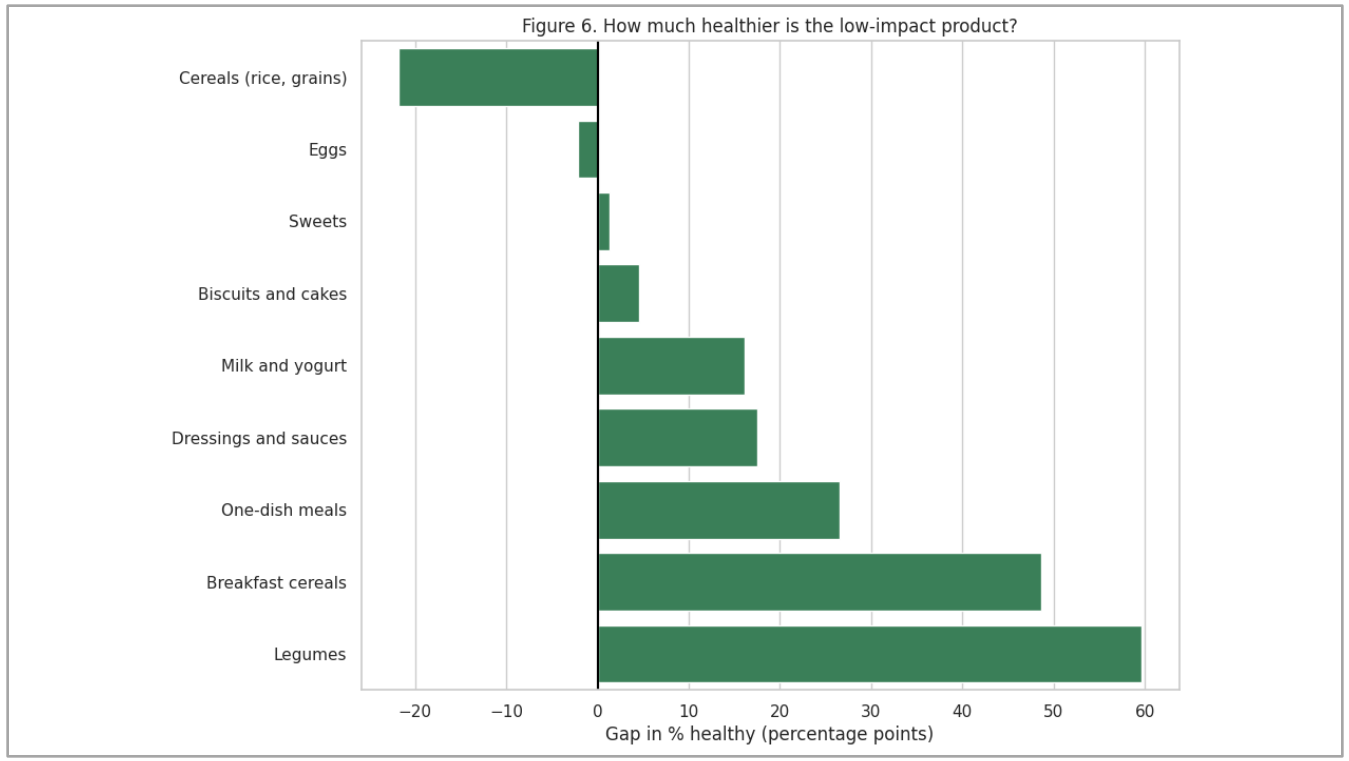


Key takeaway

The priced sample mostly mirrors the catalog, but it leans toward ready meals and appetizers (1.6×) and away from sandwiches, sweets, cheese and fish (0.3–0.7×), reflecting where our main contributor shopped. Comparing overall averages would be misleading, so every price comparison after this is made within the same food group.

The low-impact pick is healthier, and slightly cheaper

In the same food group, how much more often is the low-impact product Nutri-Score A/B?



+18 pp

more likely to be A/B than the high-impact product in the same aisle

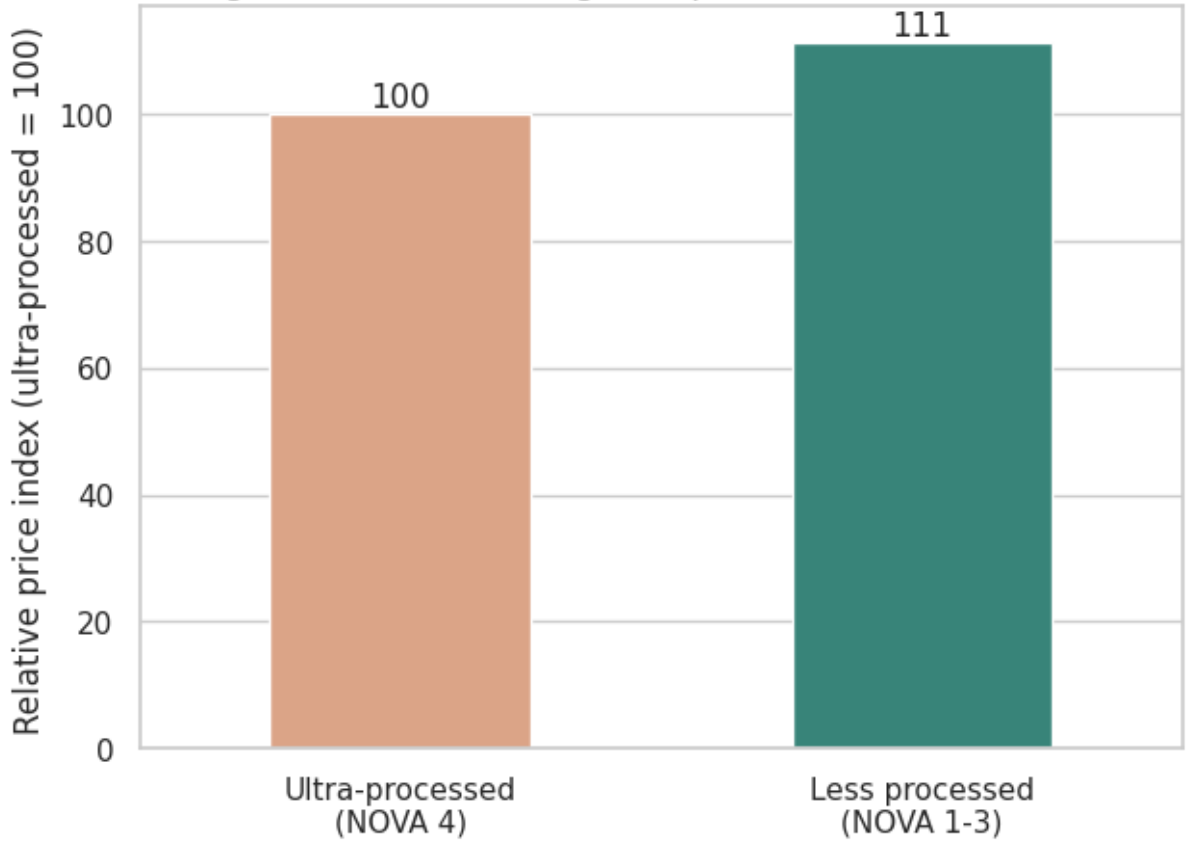
Key takeaway

Picking the low-impact product raises the chance it's healthy by about 18 percentage points on average (positive in 7 of 9 food groups), with the biggest gains in legumes (+60) and breakfast cereals (+49). It reverses for rice and grains (-22) and makes almost no difference for eggs, so the greener product is a good shortcut to the healthier one in most aisles, but not all.

Avoiding ultra-processed food raises the package price by about 11%

Per package, less-processed food costs about 11% more

Figure 10. Does avoiding ultra-processed food cost more?

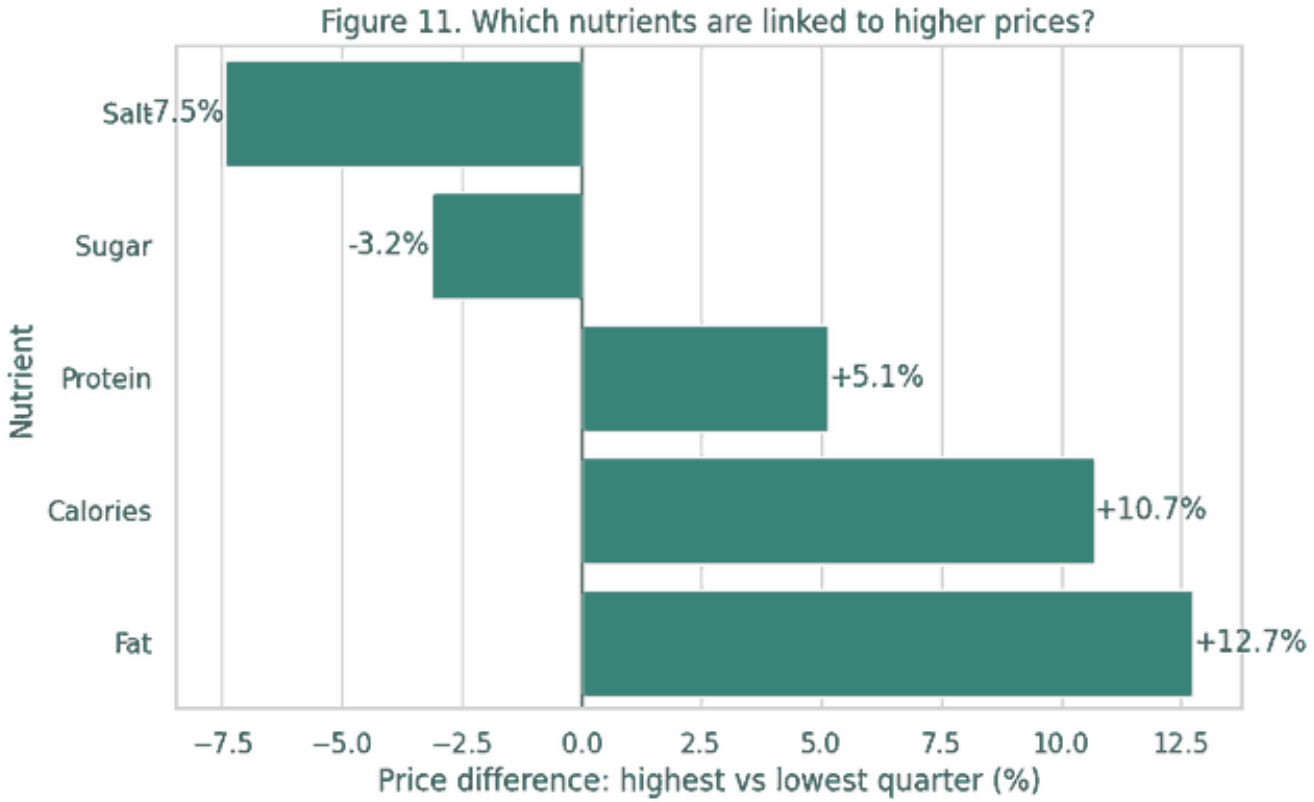


Key takeaway

Per package, less-processed products cost about 11% more than ultra-processed products in the same food group (95% CI 9–14%). This is a package price, so it mixes up the price of the food with the size of the pack, and less-processed staples like plain milk or rice often come in bigger packs. Avoiding ultra-processed food can raise the bill at the checkout, but comparing price per 100 g (Figure 9) is the fairer test of whether the food itself costs more.

Shoppers pay more for fat and calories, not for sugar or salt

Fattier, calorie-dense products cost more; sugary and salty ones cost less

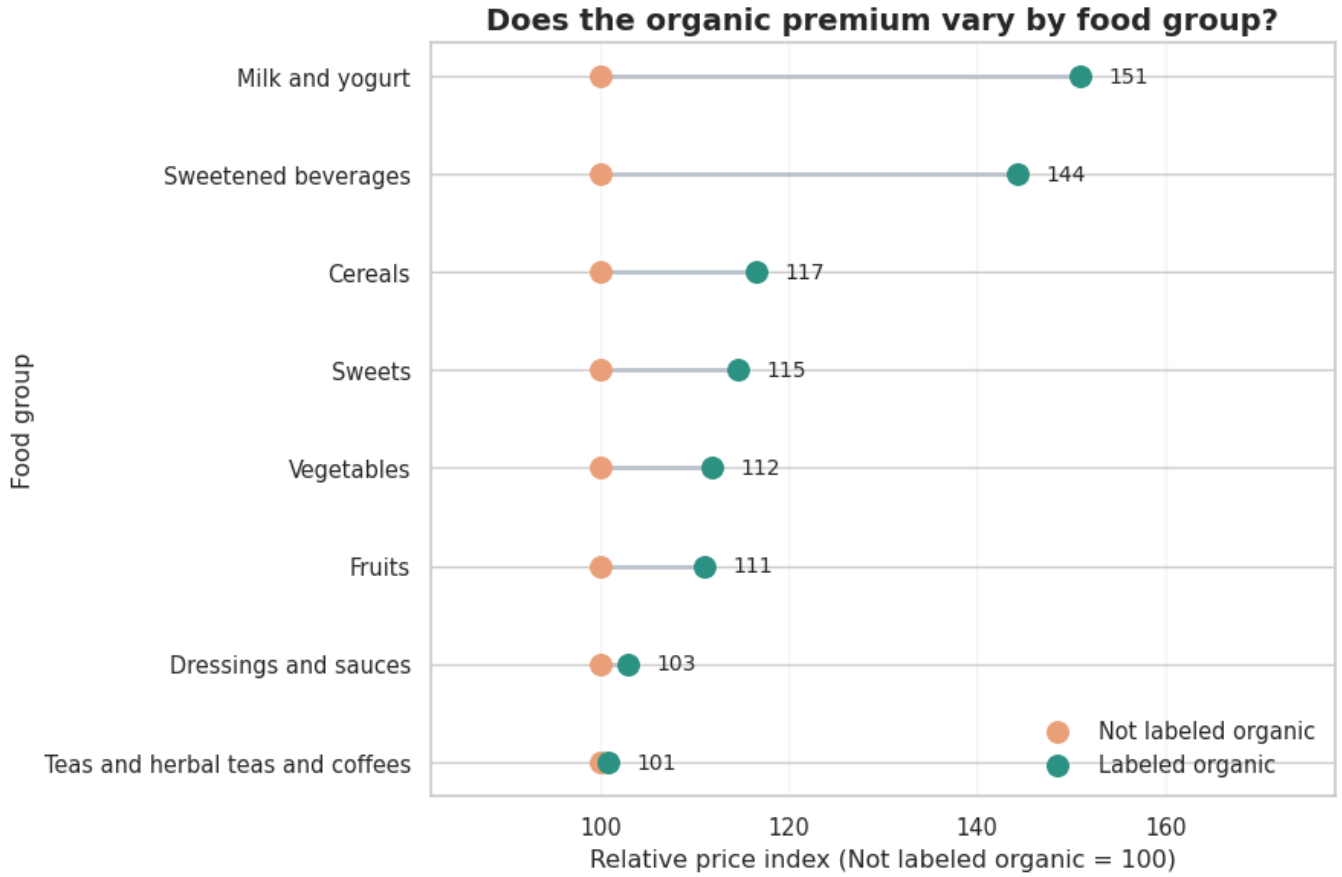


Key takeaway

Within the same food group, the fattiest and most calorie-dense products cost more per package (fat +13%, calories +11%, protein +5%), while sugary and salty products cost less (sugar −3%, salt −8%). Shoppers pay a little more for richer, more filling foods, but nothing extra for sugar or salt. Cheap sugary and salty products can make less healthy choices look like better value.

The organic premium is real, and it depends on the aisle

Organic price premium by food group



+51%

Largest shown premium: milk and yogurt (index 151; baseline 100)

Key takeaway

Organic price premiums differ by food group. Milk and yogurt (151) and sweetened beverages (144) are furthest above the not-labeled-organic baseline of 100. Teas and herbal teas and coffees (101) and dressings and sauces (103) are close to that baseline. Compare within the aisle rather than assuming one fixed premium.

Organic items were identified from product names, so some may be missed.

In most aisles, the healthier product is also the greener one

Inside each food group, are the healthier products also the greener ones?

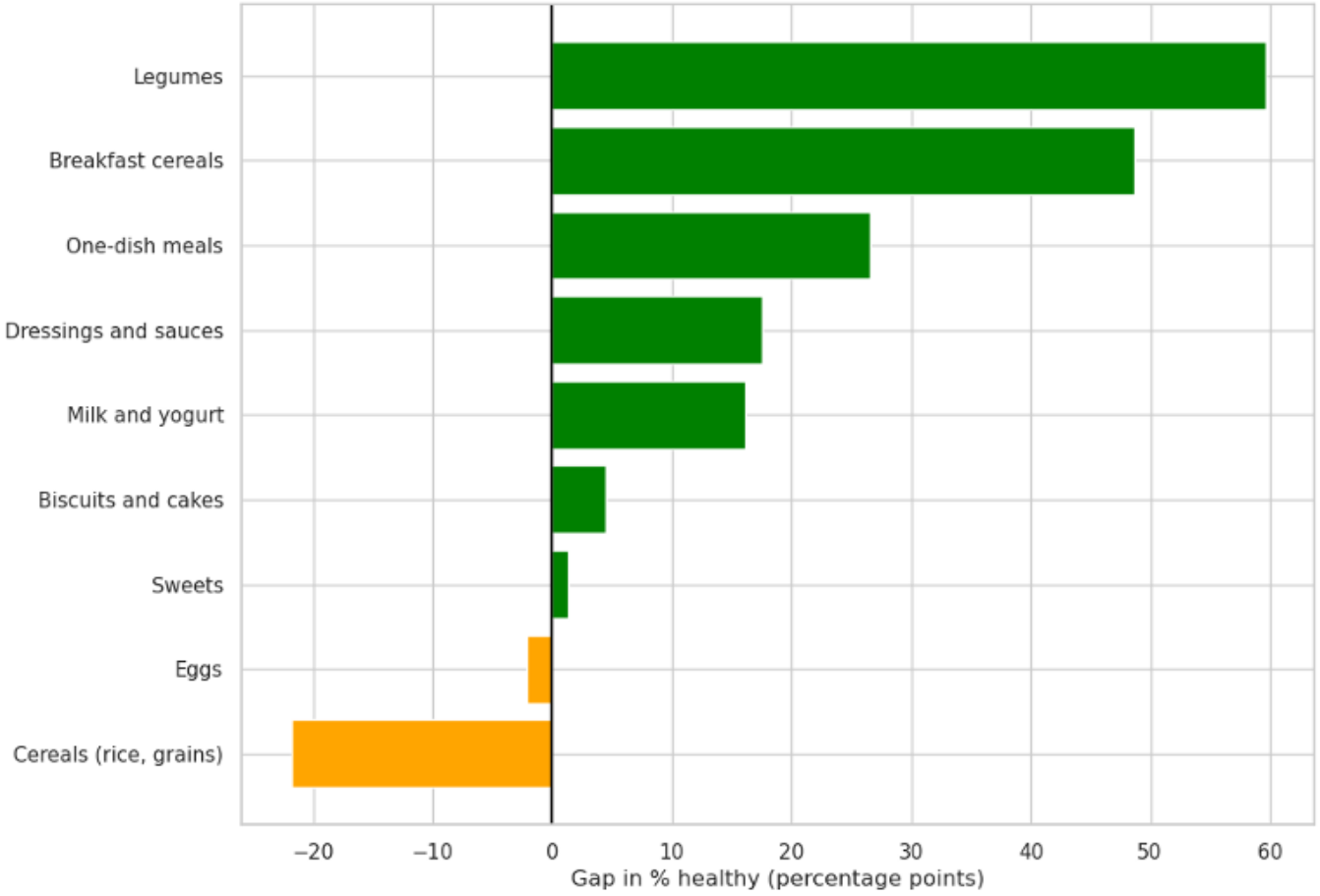


Figure 6: Gain in % Nutri-Score A/B, low- vs high-impact products, same food group

+18 pts

Within the same aisle, picking the low-impact product raises the chance it scores Nutri-Score A/B by about 18 percentage points on average (positive in 7 of 9 food groups).

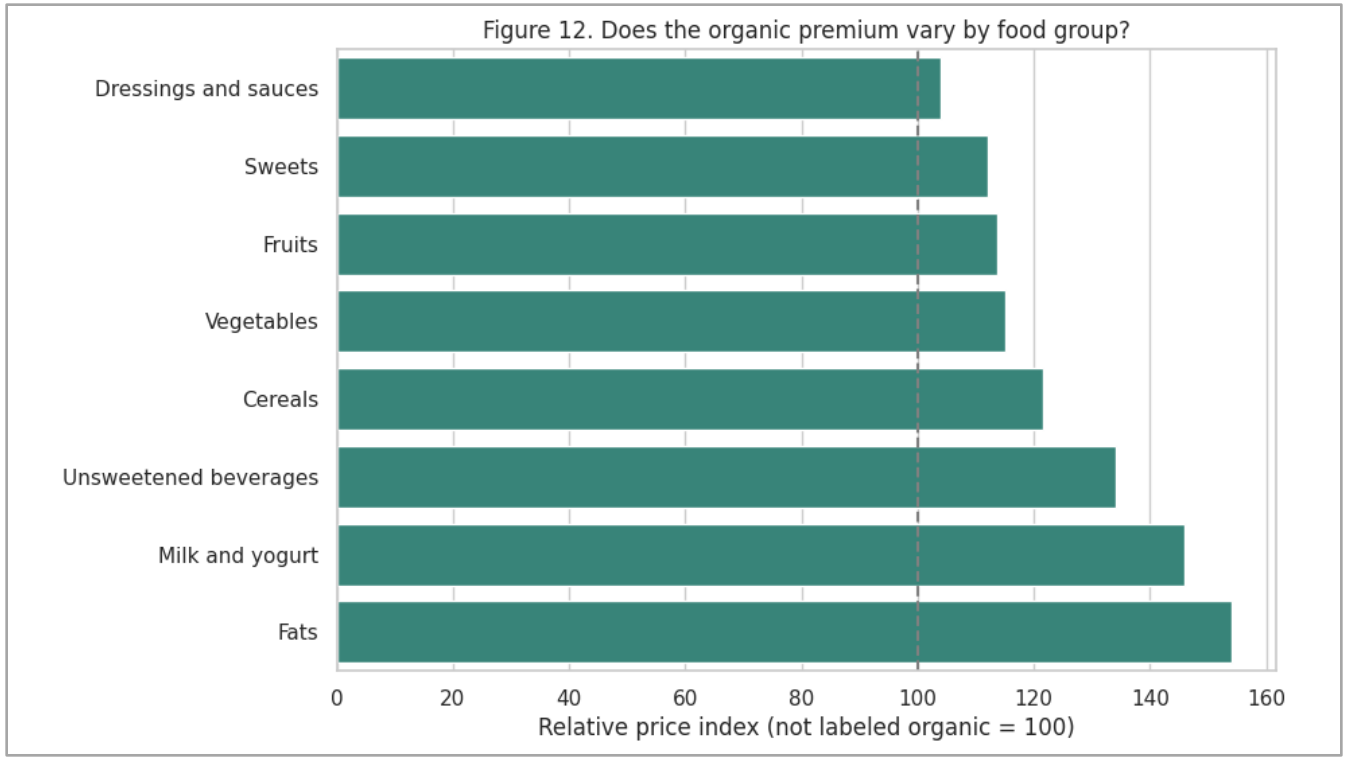
- **Biggest gains:** legumes (+60, beans vs peanut butter) and breakfast cereals (+49, plain oats vs sugary cereal)
- **Reverses:** rice and grains (−22)
- **No real difference:** eggs (−2)

Rule 2 · Pick the greener option

Therefore, Greener product is a good shortcut to the healthier one in most aisles, but not all.

The organic premium is real, and it depends on the aisle

Organic price premium by food group



+18%

organic vs non-organic per package, same food group (95% CI 15–22%)

Key takeaway

Organic products cost about 18% more per package than non-organic products in the same food group (95% CI 15–22%), and the premium varies a lot by aisle. It's small for sauces (+4%) and sweets (+12%), but large for milk and yogurt (+46%) and fats (+54%). Organic is a real premium in these stores, so shoppers on a budget should pick their organic buys aisle by aisle, starting where the premium is smallest.